

Comparison of Various Wireless Technologies

Specification	Frequency band	Bit rate	Distance	Modulation	Standardisation
Wireless USB Specification Rev. 1.1	3.1 GHz-10.6 GHz	53-480 Mbit/s	3-10 m	MB-OFDM	Sep-10
Bluetooth 4.0	2.4 GHz	1 Mbit/s	1-100 m depending on output	MB-OFDM	Jun-10
Wi-Fi (IEEE 802.11n)	2.4 GHz and/or 5 GHz	Max 450 Mbit/s per band	100 m	DSSS, DBPSK, DQPSK, CCK, OFDM	Sep-09
Wi-Fi (IEEE 802.11ac)	5 GHz	Max 6.93 Gbit/s per band	Unknown	OFDM	Pre-standard
Bluetooth 2.1 + EDR	2.4 GHz	Max 3 Mbit/s	1-100 m depending on output	GFSK	Jul-07
ZigBee (IEEE 802.15.4)	2.4 GHz	Max 250 kbit/s	80 m + mesh	DSSS	May-03
WiMAX (802.16)	2 GHz to 11 GHz	Max 70 Mbit/s	50 km	BPSK, QPSK, 16-, 64-, 256-QAM	NA
UMTS	2100 MHz Downlink and 1900 MHz Uplink	100 Mbit/s down and 50 Mbit/s up with LTE	Unknown	QPSK	NA
NFC	13.56 MHz	424 kbit/s	20 cm	ASK	NA

plementation.

Major OS providers have also started to integrate a native Bluetooth Smart stack into their updates so that cool 'accessories' (e.g., a smartphone app for a Bluetooth connected accessory) can be developed. We have a special session on this by Nitin Gupta, lead engineer, µEnergy Applications, CSR India Pvt Ltd, at EFX's Electronics Rocks 2013 conference in Bengaluru (www.bangalore.electronic rocks.com).

If the use-case requires higher data rates than what ZigBee provides, Bluetooth is the technology to go for. It provides a good balance between data rates and range, and the latest version called Bluetooth Smart (Bluetooth 4.0) is designed for such use-cases. Classic Bluetooth has a 100-metre range but also takes 100 ms to wake up from sleep, while Bluetooth Smart is limited to 50 metres but takes just 6 ms to wake up.

"Broadcom's new Wireless Internet Connectivity for Embedded Devices (WICED) platform simplifies development and implementation of Wi-Fi and Bluetooth Smart in products designed for the Internet of Things," explains Brian Bedrosian, senior director, embedded wireless, Broadcom Corporation.

The best fit. Products that benefit from lower power/voltage operation, a much longer battery life (years not

months), are non line-of-sight and have a two-way operation, benefit the most from Bluetooth Smart. More importantly, its ability to connect with a large number of nodes is very helpful in IoT implementations. It is also a technology that is compatible with most consumer electronic devices right from the box.

Alternatives.

1. *Want LAN technology rather than PAN?* If so, ZigBee is a pretty good option.

2. *Looking for something in the sub-gigahertz band?* Z-wave is a good option here, as it does most of what ZigBee does but uses the 900MHz band.

Z-Wave

Z-Wave is an alternative that suits the residential control and automation market. It has a maximum range of 30 metres, a bandwidth of 40 kbit/s and operates with low latency.

The notable point about Z-Wave is that because it operates in the sub-gigahertz band, it does not interfere with Bluetooth or Wi-Fi, though it might interfere with cordless phones.

The best fit. Z-Wave is a good option if you are looking at a low-power RF radio embedded or retrofitted into home electronics like lighting, home access control, entertainment systems and household appliances. The Z-Wave wireless protocol is optimised for reliable, low-latency

communication of small data packets and allows data rates of up to 100 kbps.

Alternatives.

1. *Need a higher data rate?* If the use-case requires higher data rates than what Z-Wave provides, Bluetooth Smart is the technology to go for.

2. *Interference from cordless phones and remotes?* Go for ZigBee, which uses the 2.4GHz band.

Wireless USB

Wireless USB is a technology that is targeted at peripherals and similar electronic devices. It operates in the 3.1 to 10.6GHz range and can send 480 Mbit/s up to 3 metres and 110 Mbit/s up to 10 metres. The wireless transmission techniques used in Wireless USB are very robust, as they do not interfere with or get interfered by other technologies like Bluetooth and Wi-Fi. This is due to the use of a wide frequency spectrum with extremely low power at any one given frequency and a high-power spectral density.

The best fit. The ultra-wideband signals used in Wireless USB will not be affected by nearby devices running Wi-Fi, Bluetooth, or even by cordless phones. This makes it an ideal technology to be used in the living room or work environment, where a multitude of alternate wireless devices are also in use.